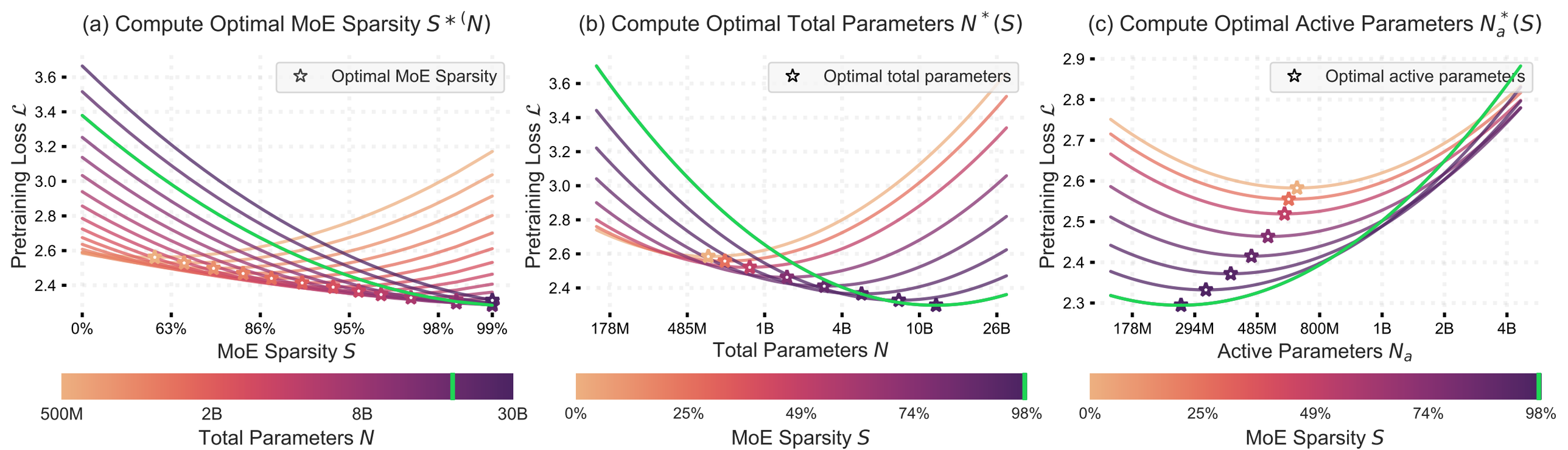
(a) Compute Optimal MoE Sparsity $S*(N)$
Optimal MoE Sparsity
Pretraining Loss $\mathcal{L}$
3.6
3.4
3.2
3.0
2.8
2.6
2.4
0%
63%
86%
95%
98%
99%
MoE Sparsity S
500M
2B
8B
30B
Total Parameters N
(b) Compute Optimal Total Parameters $N^*(S)$
Optimal total parameters
Pretraining Loss $\mathcal{L}$
3.6
3.4
3.2
3.0
2.8
2.6
2.4
178M
485M
1B
4B
10B
26B
Total Parameters N
0%
25%
49%
74%
98%
MoE Sparsity S
(c) Compute Optimal Active Parameters $N_a^*(S)$
Optimal active parameters
Pretraining Loss $\mathcal{L}$
2.9
2.8
2.7
2.6
2.5
2.4
2.3
178M
294M
485M
800M
1B
2B
4B
Active Parameters N_a
0%
25%
49%
74%
98%
MoE Sparsity S